

LOEI AGROMET, Thailand

WMO: 483500

Lat: 17.4094N

Lon: 101.7294E

Elev: 264

StdP: 98.20

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.6	13.6	9.2	7.5	15.2	10.9	8.4	15.9	4.3	26.4	3.7	26.0	0.1	N/A	0.261

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.2	38.0	25.9	36.5	25.6	35.2	25.8	27.9	33.5	27.6	33.0	27.2	32.4	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.8	31.2	26.2	22.3	30.7	25.8	21.8	30.2	91.4	33.6	89.8	33.3	88.1	32.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.9	3.3	2.9	DB	8.4	39.4	2.7	1.6	6.5	40.5	4.9	41.5	3.4	42.3	1.5	43.5	(4)
(5)			WB	8.3	28.9	2.5	0.6	6.5	29.4	5.0	29.7	3.6	30.0	1.7	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	22.4	24.8	26.9	29.1	28.5	28.5	28.0	27.6	27.4	26.5	25.0	22.3
	DBStd	2.96	2.58	2.41	2.56	2.21	1.62	1.27	1.45	1.23	1.04	1.43	1.94	2.72
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	11	4	1	1	0	0	0	0	0	0	0	0	4
	CDD10.0	5993	384	413	524	572	573	556	558	546	523	511	449	382
	CDD18.3	2962	130	181	267	322	314	306	300	288	273	253	199	128
	CDH23.3	27479	939	1693	2802	3640	3062	3212	2942	2573	2306	1940	1467	905
CDH26.7	11604	297	788	1486	1997	1345	1327	1103	904	806	727	544	279	
Wind	WSAvg	0.6	0.6	0.8	0.8	0.8	0.6	0.6	0.6	0.6	0.5	0.5	0.5	0.6
Precipitation	PrecAvg	1218	6	14	38	93	203	148	137	189	249	111	17	4
	PrecMax	1543	43	53	67	178	389	339	196	351	528	211	54	40
	PrecMin	1008	0	0	1	20	78	49	52	103	131	20	1	0
	PrecStd	155	13	17	20	48	81	74	43	74	116	61	18	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	36.2	39.0	40.6	37.9	36.1	37.0	34.6	34.1	34.4	33.7	32.6
		MCWB	23.0	23.7	25.2	26.5	27.4	27.0	26.6	27.3	27.6	26.7	26.0	24.1
	2%	DB	31.1	34.6	37.2	38.8	35.9	34.6	34.6	33.2	33.1	32.8	32.1	31.1
		MCWB	22.5	23.6	24.3	26.3	27.3	27.0	26.8	26.9	27.3	26.0	25.1	23.8
	5%	DB	29.9	33.3	35.8	37.1	34.3	33.5	33.0	32.1	32.0	31.7	31.1	29.6
		MCWB	22.3	23.3	24.3	26.0	27.0	26.9	26.7	26.6	27.0	25.9	24.8	23.1
	10%	DB	28.4	31.6	33.9	35.4	32.8	32.2	31.5	31.1	31.0	30.6	30.0	28.1
		MCWB	21.6	22.9	24.1	25.8	26.8	26.7	26.4	26.4	26.6	25.5	24.2	22.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	25.6	26.6	28.0	28.5	28.2	28.0	28.1	28.2	27.4	26.7	25.6
		MCDB	30.1	33.1	36.0	37.2	35.1	33.4	33.1	32.8	32.7	32.1	31.7	30.6
	2%	WB	23.8	24.6	25.7	27.2	27.7	27.6	27.4	27.4	27.7	26.8	26.0	24.3
		MCDB	28.8	32.2	34.2	35.8	33.9	32.9	32.3	31.9	31.9	31.3	30.8	29.5
	5%	WB	23.0	24.0	25.1	26.7	27.3	27.2	27.0	26.9	27.2	26.3	25.2	23.5
		MCDB	27.7	31.5	33.3	34.9	33.1	32.2	31.7	31.2	31.2	30.6	29.9	28.2
	10%	WB	22.2	23.3	24.5	26.1	26.8	26.8	26.5	26.5	26.7	25.8	24.6	22.7
		MCDB	26.6	30.1	32.3	33.9	32.0	31.4	30.8	30.4	30.4	29.7	28.7	27.0
Mean Daily Temperature Range	5% DB	MDBR	12.2	14.2	13.3	12.2	9.2	8.0	7.4	7.3	7.6	8.5	10.3	11.7
		MCDBR	14.0	16.4	16.8	14.9	11.6	9.8	9.9	9.0	9.2	10.2	11.3	12.7
		MCWBR	6.2	6.3	6.0	5.0	4.1	3.5	3.8	3.6	4.1	4.2	5.0	5.7
	5% WB	MCDBR	11.4	14.2	14.0	12.8	10.2	8.7	8.4	8.2	8.4	9.0	10.0	11.5
		MCWBR	5.3	5.7	5.3	4.8	4.1	3.5	3.6	3.6	4.1	4.2	4.7	5.5
Clear-Sky Solar Irradiance	taub	0.466	0.571	0.670	0.623	0.494	0.444	0.432	0.447	0.494	0.538	0.453	0.456	
	taud	2.110	1.852	1.660	1.809	2.191	2.358	2.387	2.324	2.178	2.011	2.224	2.157	
	Ebn at Noon	813	744	688	725	814	847	859	853	815	763	821	810	
	Edh at Noon	151	205	256	221	149	125	122	130	150	172	134	141	
All-Sky Solar Radiation	RadAvg	4.65	5.09	5.14	5.51	5.18	4.87	4.55	4.41	4.50	4.70	4.83	4.51	
	RadStd	0.31	0.23	0.33	0.30	0.36	0.35	0.31	0.28	0.28	0.32	0.20	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+116

Nomenclature: See separate page